

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Distributed System**

Code No: **CACS352** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Why is a distributed operating system needed? Differentiate between network operating systems and distributed operating systems.
3. How does virtualization contribute to the functionality of a distributed system?
4. What is mutual exclusion and how is it maintained in distributed environments?
5. What is code migration? Explain the migration models.
6. Define name, address, and identifier. Explain the types of communication in a distributed system.
7. Explain the different aspects of distribution transparency in brief.
8. Write short notes on (any two): a. Public key cryptography b. Security Management c. RPC

Group C

Attempt any TWO questions[2x10=20]

9. What is clock synchronization? Why is clock synchronization needed in a distributed system? Explain the physical clock synchronization algorithm in detail.
10. What are the types of failures that may occur in a distributed system? Explain the methods used to recover from the crash? Explain the one-phase commit and two-phase commit protocol in detail.
11. What is replication? Explain data-centric consistency models with a suitable example. Explain the types of name resolution algorithms used in distributed systems.